

WELCOME

CSC380 - Principles of Data Science

TODAY'S PLAN

- Syllabus overview: Check complete syllabus at D2L. Send me an email if any part is not clear.
- Introduction to Probability

COURSE INSTRUCTOR

- Cesim Erten, PhD from University of Arizona in CS (2004)
- Research:
Bioinformatics, Computational network biology, Graph algorithms
- Teaching
Data Science, Artificial Intelligence, Algorithms, Automata Theory
Formal Languages, Data Structures, Bioinformatics ...
- Office Hours: TBA

COURSE TAs

- Yao Zhao, Ruoshan Lan, Zachary Hansen, Desmond Thomas
Goodman-Ahearn, Advait Khopade

COURSE COMMUNICATIONS

- Outside-lecture communications will be through Piazza.
Signup link: In D2L
- Assignment/Project submissions through gradescope.
Signup code: In D2L
- The instructor and TAs can also be reached through email.
- Come to instructor/TA office hours.

COURSE MATERIALS

- Textbooks:

- Watkins, J., "An Introduction to the Science of Statistics: From Theory to Implementation"
- Wasserman, L. "All of Statistics: A Concise Course in Statistical Inference."
- James, G. et al. "An Introduction to Statistical Learning with Applications in Python"
- Steven S. Skiena, "The Data Science Design Manual"
- Lau, S. et al. "Learning Data Science"

- Lecture slides will be shared through D2L.

GRADING ITEMS

- Assignments (5% each):
 - Problem solving + Coding. Must be done individually.
 - 7 assignments (lowest grade dropped)
- Project (16% total)
 - Can be done in pairs
- Quizzes (1% each)
 - 12 quizzes (lowest two dropped). One question in each.
 - Announced one day before. Previous 2 lectures. 5 mins.
- Exams (44% total)
 - 1 midterm (20%) + final (24%)

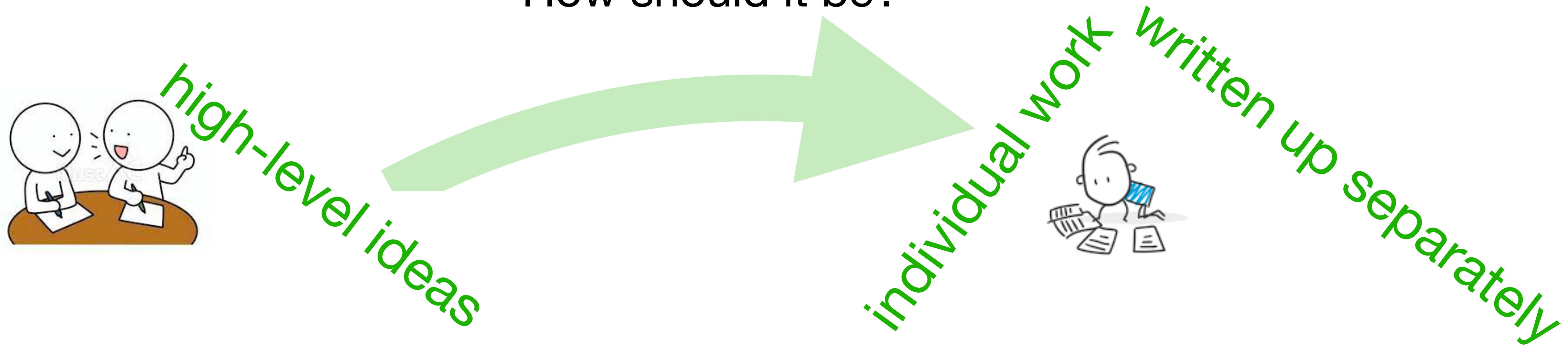
BONUS OPPORTUNITIES

- **5% bonus:** All conditions below must be satisfied.
 - ≥ 4 “participation instances” in lectures
 - ≥ 4 “participation instances” in piazza
 - ≥ 4 appearances in instructor’s office hours
 - ≥ 4 SI session participations
- “participation instance” in lecture: **PAIR UP & DISCUSS** 
Answer a group (with neighbor) discussion question correctly
- “participation instance” in piazza:
A correct answer to a technical question (at most one/month counted)

MORE ON ASSIGNMENTS

- Use of external sources: For instance, possible collaboration with friends.

How should it be?



- If you discuss high-level ideas with a friend mention his/her name in your solution.
- **Verbatim copying of solutions** from any external source (web search, ChatGPT, etc.) is **not acceptable**. If you make use of any external source mention the source in your solution and **make sure that your solution is your own work**.
- Submissions after the due date and time are **not accepted**.

FURTHER IMPORTANT POINTS

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): <https://cirt.arizona.edu/case-emergency/overview>

Also watch the video available at https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/common/learningeventdetail/crtfy0000000000003560

UNIVERSITY/DEPARTMENT POLICIES

University-wide Policies link

Links to the following UA policies are provided here,

<http://catalog.arizona.edu/syllabus-policies:>

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy

Department-wide Syllabus Policies and Resources link

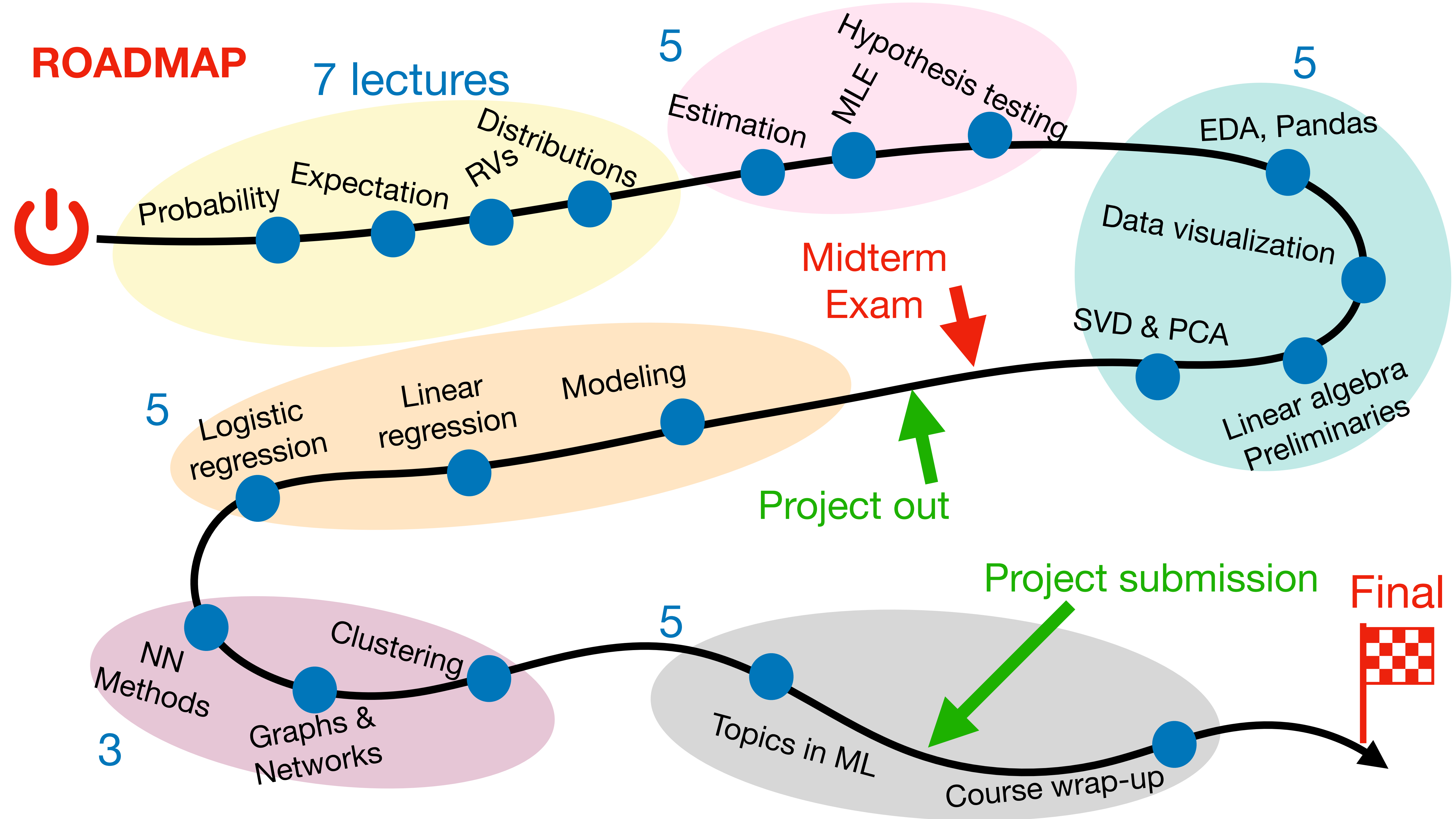
Links to the following departmental syllabus policies and resources are provided here,

<https://www.cs.arizona.edu/cs-course-syllabus-policies> :

- Department Code of Conduct
- Class Recordings
- Illnesses and Emergencies
- Obtaining Help
- Preferred Names and Pronouns
- Confidentiality of Student Records
- Additional Resources
- Land Acknowledgement Statement

ROADMAP

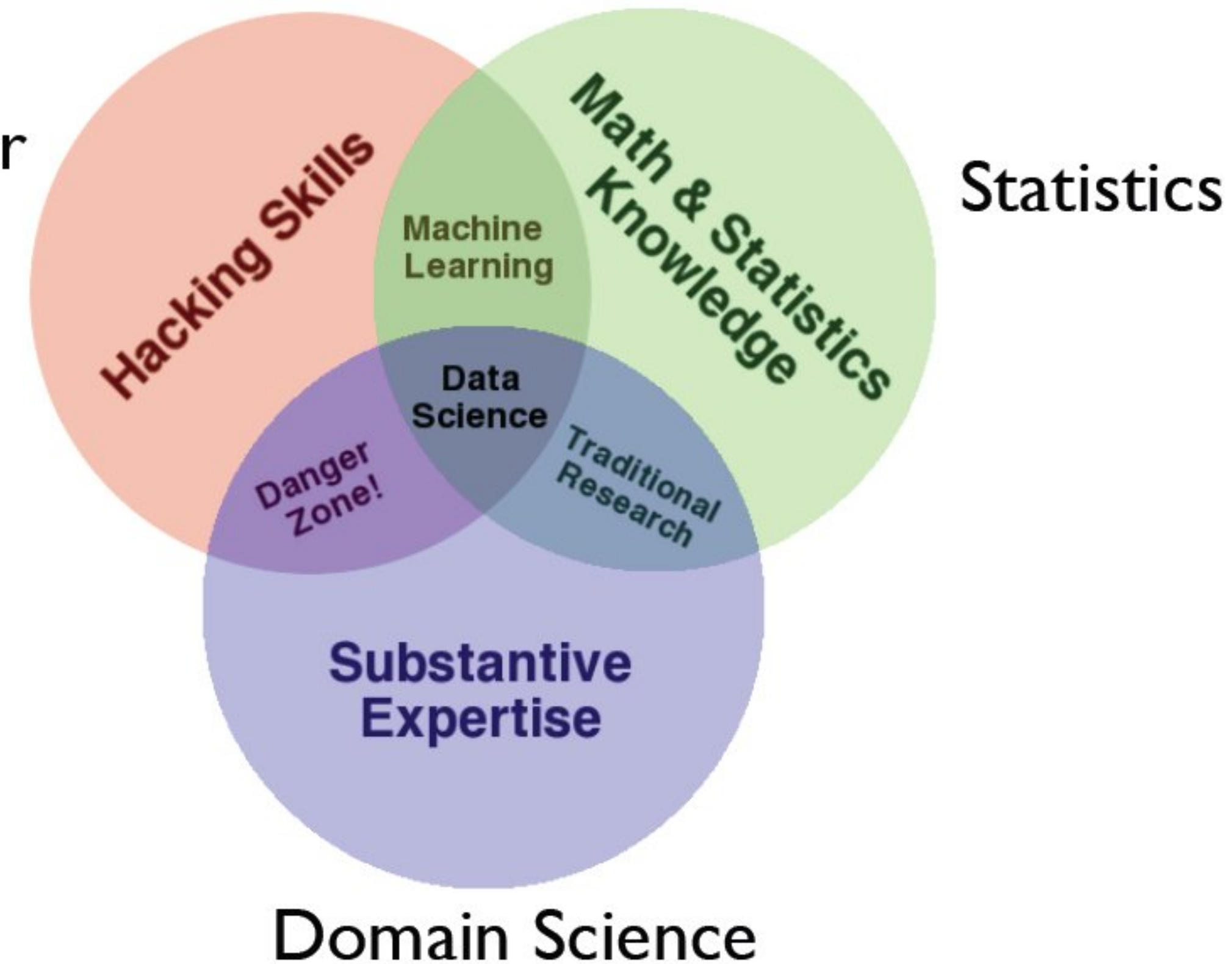
7 lectures



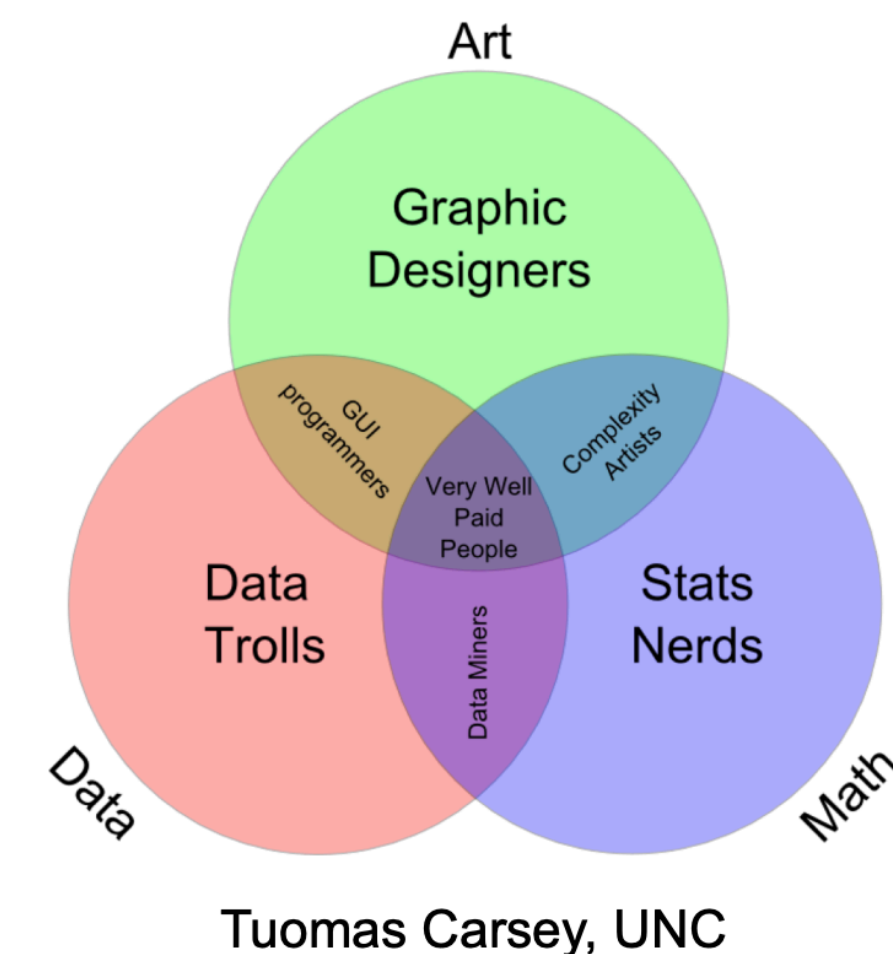
INTRODUCTION TO DATA SCIENCE

- **Broad & Interdisciplinary :**
not a single discipline, at the intersection of multiple disciplines: statistics, computer science, operations research, statistical and machine learning, data warehousing, visualization, mathematics, information science, ...
- **Insight-focused:**
find **insights** in data and leverage them to inform decision making

Computer
Science



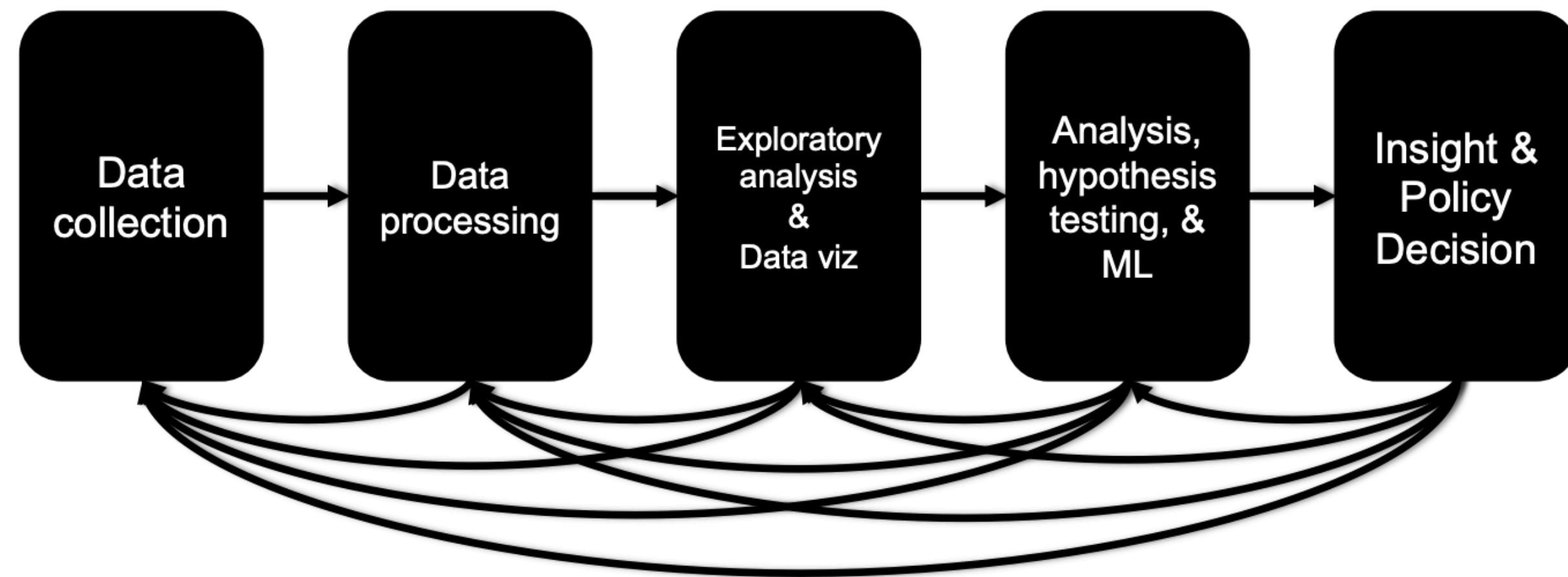
Drew Conway



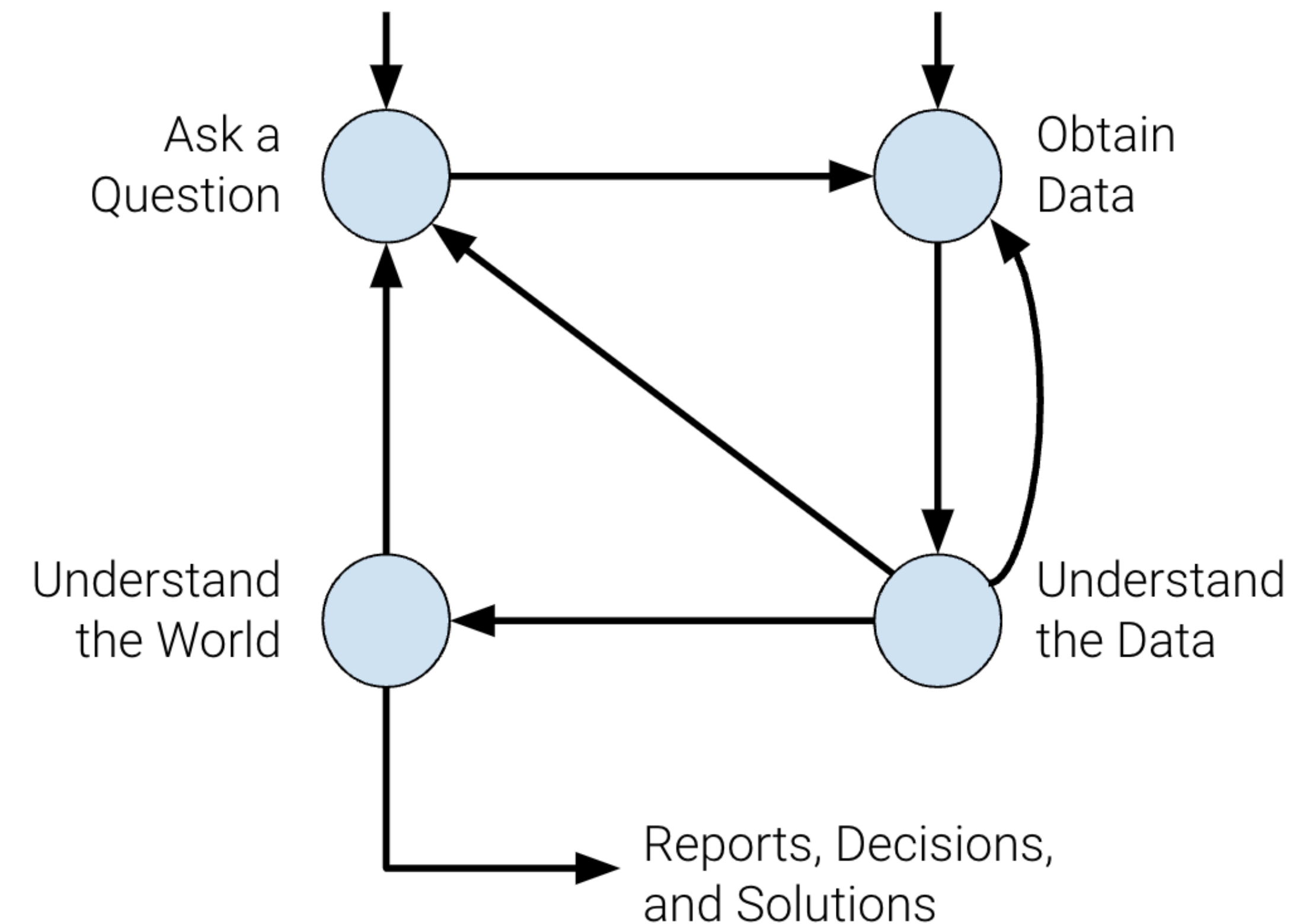
Tuomas Carsey, UNC

THE DATA SCIENCE LIFE CYCLE

Many different data science life cycles: core ideas similar



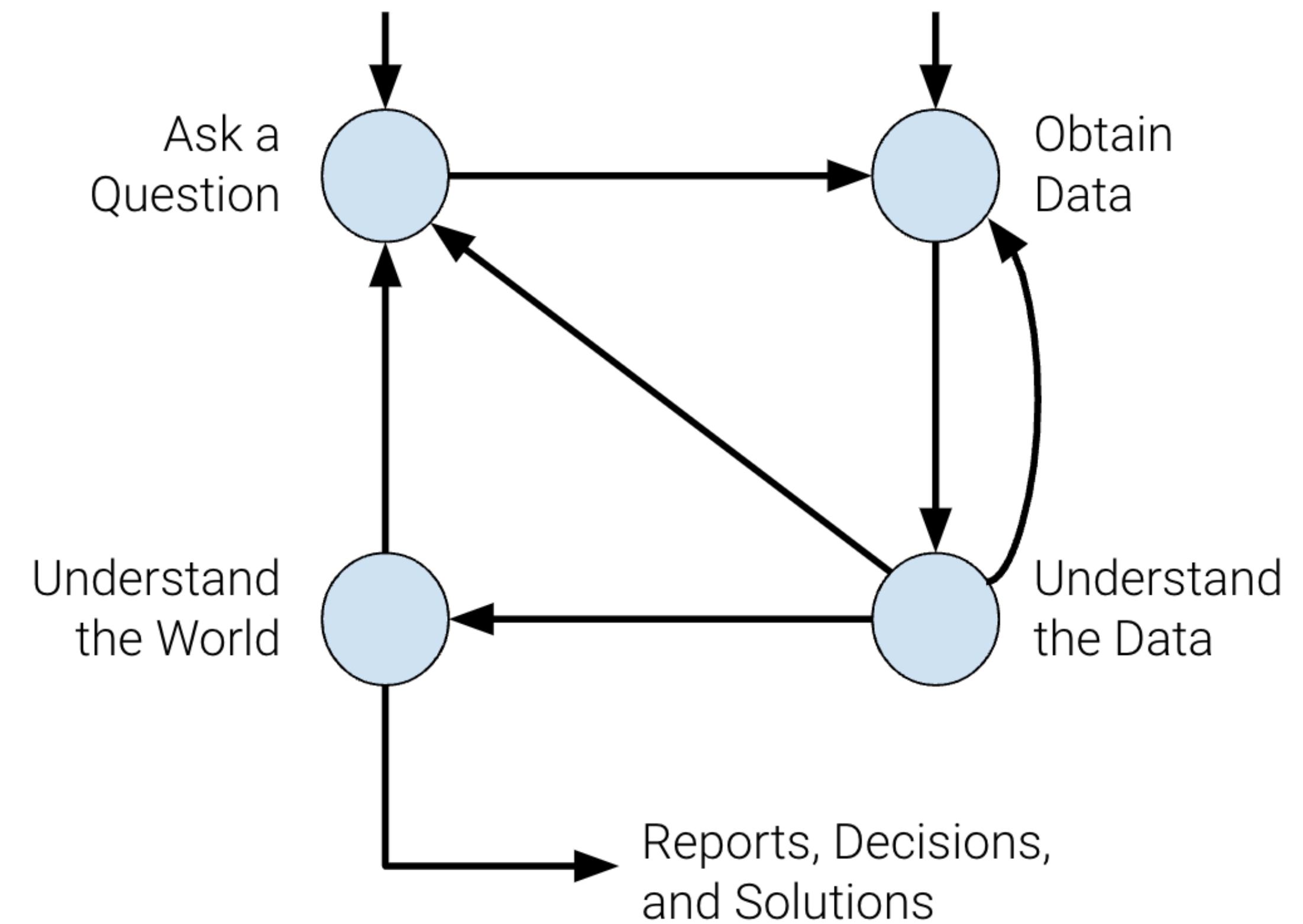
J. Dickerson, UMD



Data 100, UC Berkeley

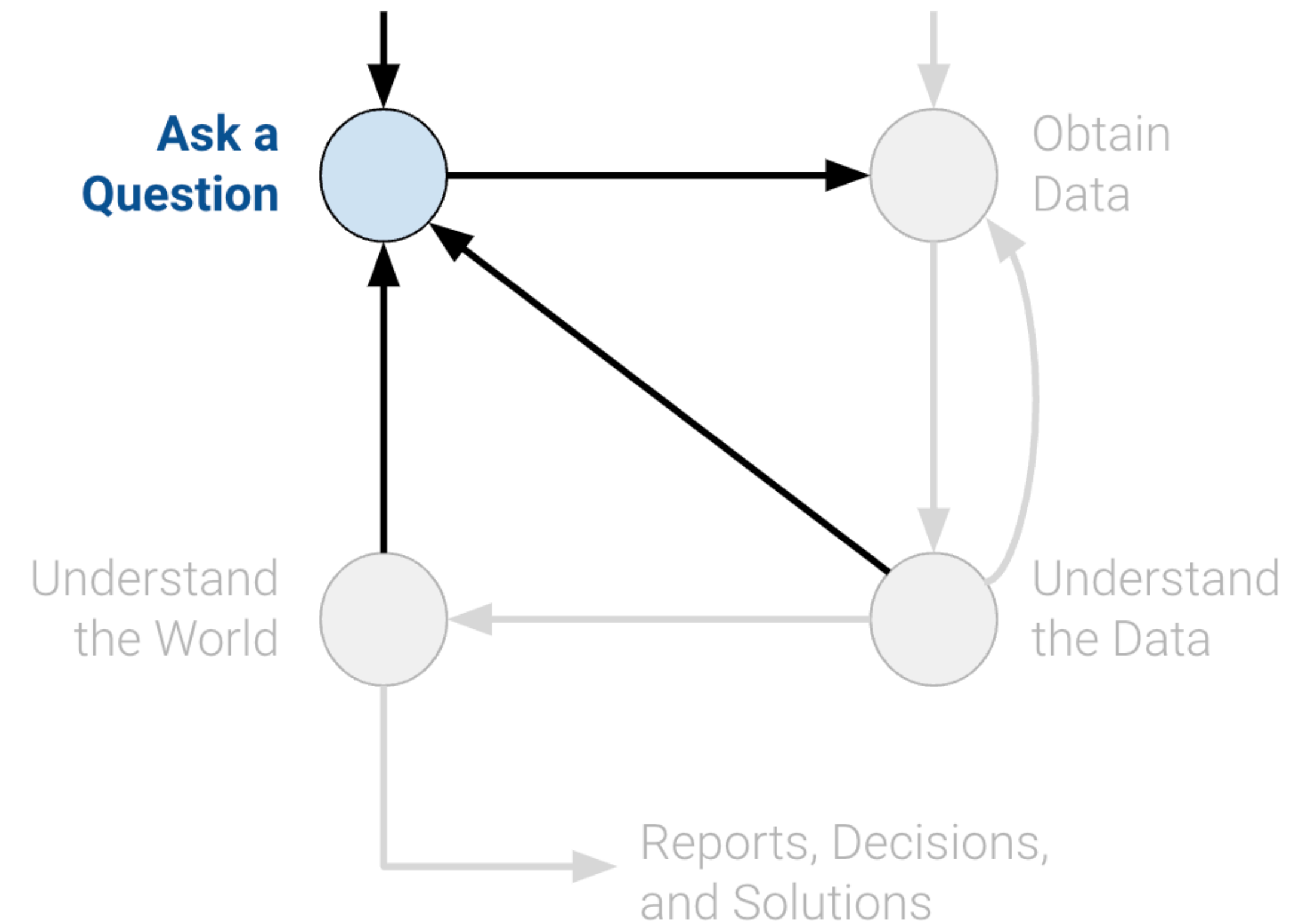
THE DATA SCIENCE LIFE CYCLE

- Many different data science life cycles: core ideas similar



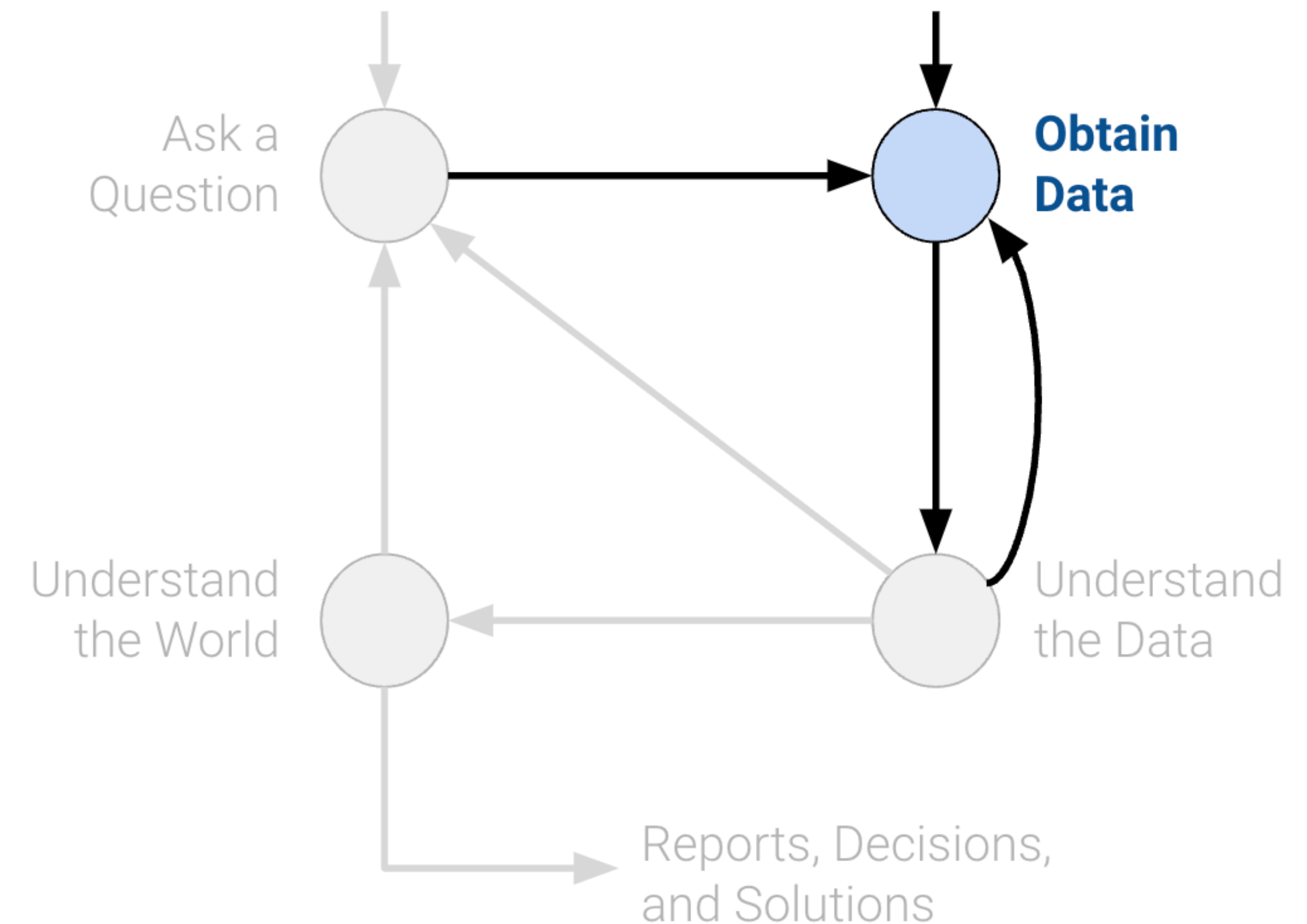
THE DATA SCIENCE LIFE CYCLE

- What do we want to know?
- What problems are we solving?
- What hypotheses are we testing?
- What are our success metrics?



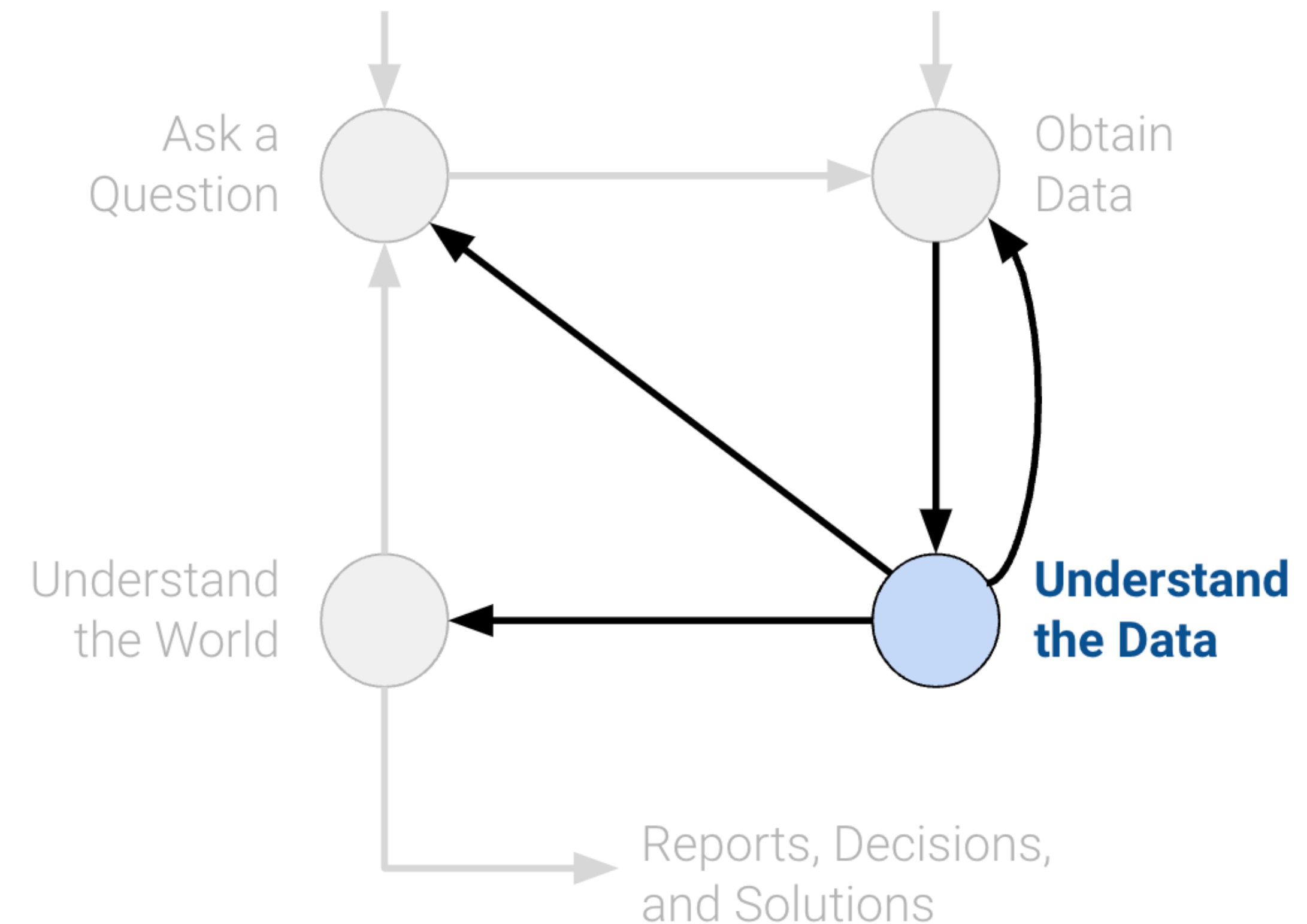
THE DATA SCIENCE LIFE CYCLE

- What data do we have and what data do we need?
- How will we sample more data?
- Is our data representative of the population we want to study?



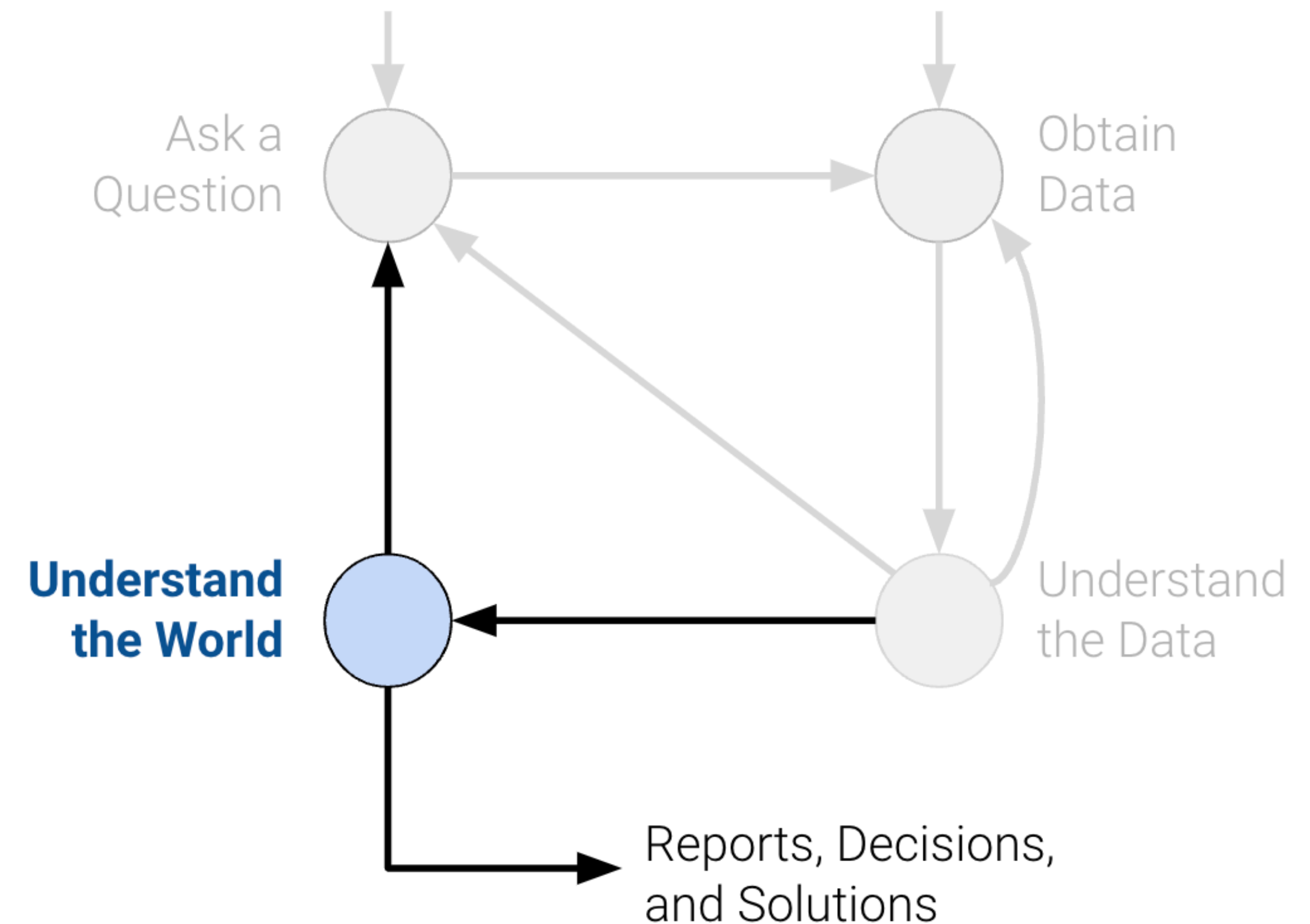
THE DATA SCIENCE LIFE CYCLE

- How is our data organized and what does it contain?
- Do we already have relevant data?
- What are the biases, anomalies, or other issues with the data?
- How do we transform the data to enable effective analysis?



THE DATA SCIENCE LIFE CYCLE

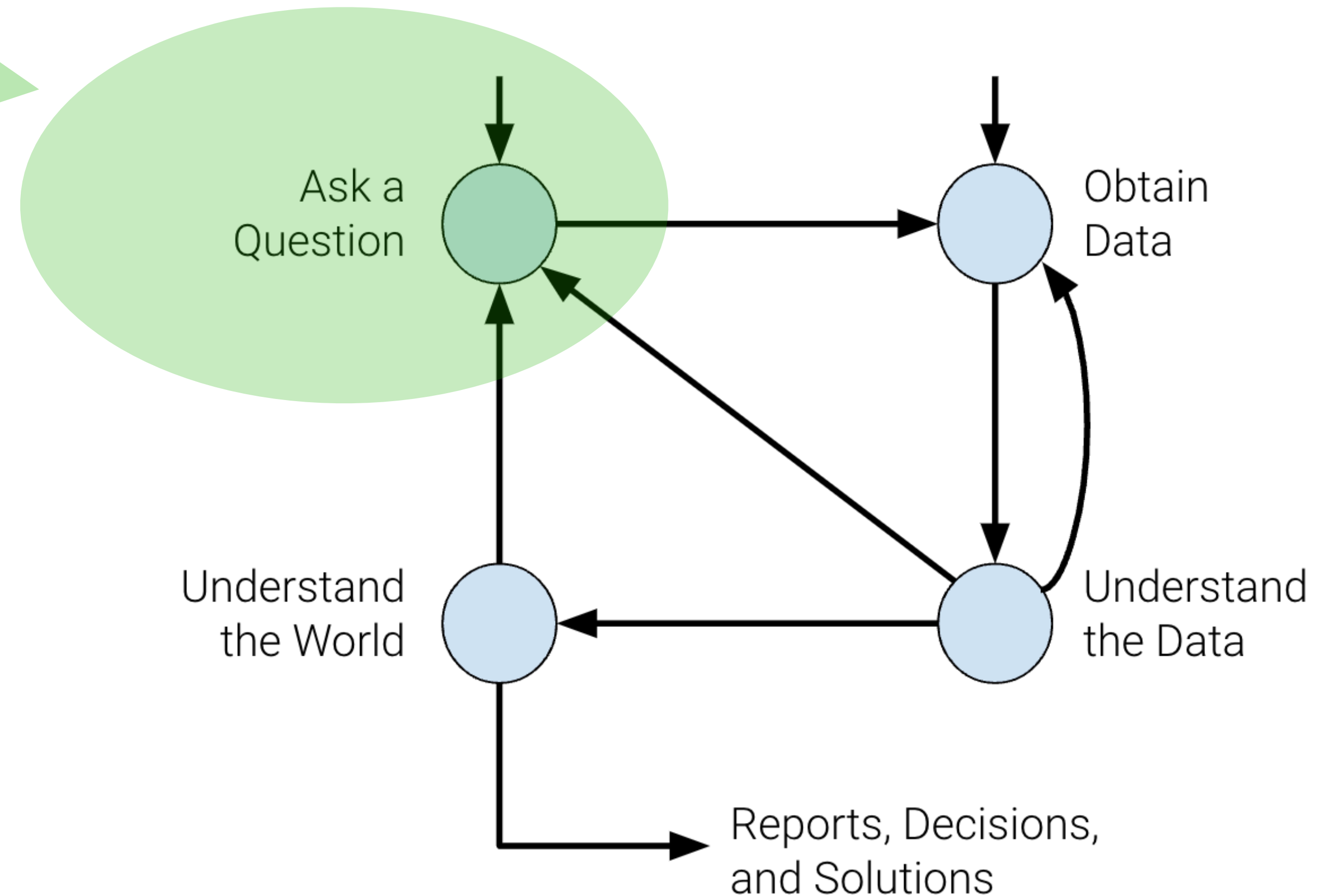
- What does the data say about the world?
- Does it answer our questions or accurately solve the problem?
- How robust are our conclusions and can we trust the predictions?



THE DATA SCIENCE LIFE CYCLE

- Many different data science life cycles: core ideas similar

Let's focus on this



ASKING QUESTIONS

- The IMDb movie database: Movies

imdb.com/title/tt0133093/?ref_=chttp_i_16

The Matrix

1999 · R · 2h 16m

+ EANU REEVES LAURENCE FISHBURNE





Action

Sci-Fi

When a beautiful stranger leads computer hacker Neo to a forbidding underworld, he discovers the shocking truth--the life he knows is the elaborate deception of an evil cyber-intelligence.

Directors

Lana Wachowski · Lilly Wachowski

Writers

Lilly Wachowski · Lana Wachowski

IMDb Rating

★ 8.7 / 10

2M

Streaming



PRIME VIDEO CH

+ Add to

Added to

Screenshot

imdb.com/title/tt0133093/?ref_=chttp_i_16

Details

Edit

Release date

March 31, 1999 (United States)

>

Countries of origin

United States · Australia

Official sites

HBO Max (United States) · Official Facebook

>

Language

English

Also known as

Ma Trận

>

Filming locations

Nashville, Tennessee, USA (exterior scenes: skyline in opening Trinity rooftop chase)

>

Production companies

Warner Bros. · Village Roadshow Pictures · Groucho Film Partnership

>

See more company credits at IMDbPro

Box office

Edit

Budget

\$63,000,000 (estimated)

Opening weekend US & Canada

\$27,788,331 · Apr 4, 1999

Gross US & Canada

\$172,076,928

Gross worldwide

\$467,222,728

Screenshot

ASKING QUESTIONS

- The IMDb movie database: Movies



Keanu Reeves

Biography

Jump to ▾

Overview

Born

September 2, 1964 · Beirut, Lebanon

Birth name

Keanu Charles Reeves

Nicknames

The Wall · The One

Height

6' 1" (1.86 m)

Family

Children

No Children

Parents

Samuel Nowlin Reeves
Patric Reeves

Relatives

Kim Reeves (Sibling)
Karina Miller (Half Sibling)
Emma Reeves (Half Sibling)

Trademarks

Intense contemplative gaze

Deep husky voice

Known for playing stoic reserved characters

Friendly, down-to-earth personality



The Matrix (1999)

Full Cast & Crew

IMDbPro See agents for this cast & crew on IMDbPro

Directed by

Lana Wachowski ... (as The Wachowski Brothers)
Lilly Wachowski ... (as The Wachowski Brothers)

Writing Credits (WGA)

Lilly Wachowski ... (written by) (as The Wachowski Brothers) &
Lana Wachowski ... (written by) (as The Wachowski Brothers)

Cast (in credits order) verified as complete

	Keanu Reeves	...	Neo
	Laurence Fishburne	...	Morpheus
	Carrie-Anne Moss	...	Trinity
	Hugo Weaving	...	Agent Smith
	Gloria Foster	...	Oracle
	Joe Pantoliano	...	Cypher
	Marcus Chong	...	Tank
	Julian Arahanga	...	Apoc

ASKING QUESTIONS

- Stars live longer/shorter as compared to average person?
- Can we predict the rating of a new movie?
- Can we predict the filming locations?
Which features are important for it? Genre, budget, ...?
- What is the age distribution of actors and actresses in film?
When the characters they played are couples?
Any genres showing different distribution than the overall?
- What does the social network of actresses/actors look like?

FORMAT OF DATA

WIKIPEDIA

The Free Encyclopedia

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Log in

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Relationship to statistics

Etymology

Early usage

Modern usage

Data Science and Data Analysis

History

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From Wikipedia, the free encyclopedia

Not to be confused with [information science](#).

Data science is an [interdisciplinary](#) academic field^[1] that uses [statistics](#), [scientific computing](#), [scientific methods](#), processes, [algorithms](#) and systems to extract or extrapolate [knowledge](#) and insights from potentially noisy, structured, or [unstructured data](#).^[2]

Data science also integrates domain knowledge from the underlying application domain (e.g., natural sciences, information technology, and medicine).^[3] Data science is multifaceted and can be described as a science, a research paradigm, a research method, a discipline, a workflow, and a profession.^[4]

Data science is a "concept to unify [statistics](#), [data analysis](#), [informatics](#), and their related [methods](#)" to "understand and analyze actual [phenomena](#)" with [data](#).^[5] It uses techniques and theories drawn from many fields within the context of [mathematics](#), statistics, [computer science](#), [information science](#), and [domain knowledge](#).^[6] However, data science is different from [computer science](#) and information science. [Turing Award](#) winner [Jim Gray](#) imagined data science as a "fourth paradigm" of science ([empirical](#), [theoretical](#), [computational](#), and now data-driven) and asserted that "everything about science is changing because of the impact of [information technology](#)" and the [data deluge](#).^{[7][8]}

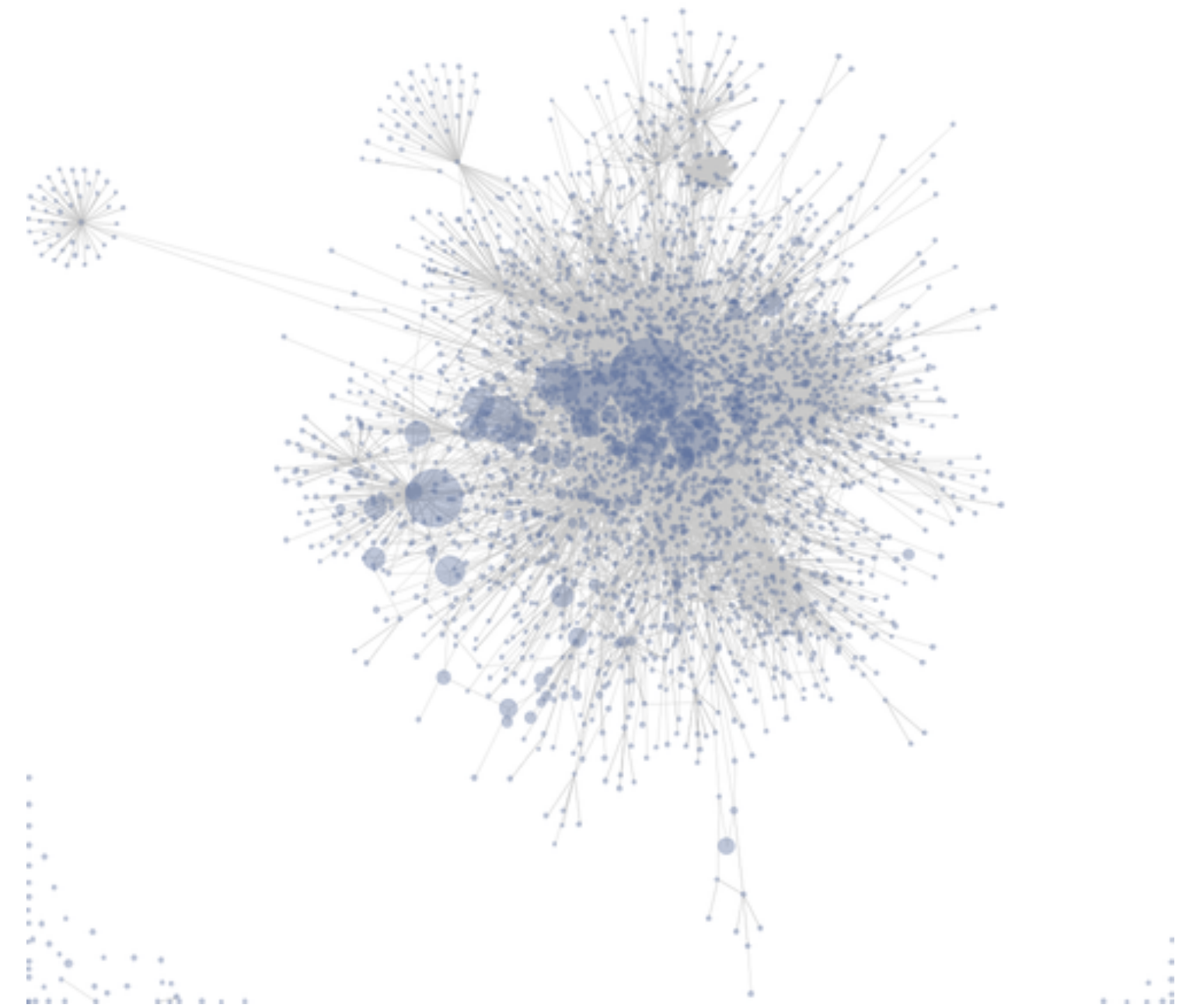
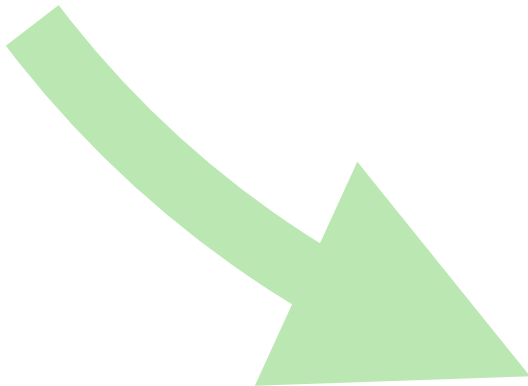
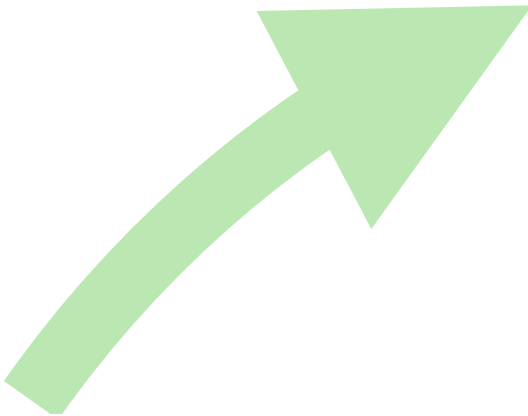
A **data scientist** is a professional who creates programming code and combines it with statistical knowledge to create insights from data.^[9]

Foundations

[edit]

Data science is an [interdisciplinary field](#)^[10] focused on [extracting knowledge](#) from typically [large data sets](#) and applying the knowledge and

The existence of [Comet NEOWISE](#) ⁵ (here depicted as a series of red dots) was discovered by analyzing [astronomical survey](#) data acquired by a [space telescope](#), the [Wide-field Infrared Survey Explorer](#).



	“interdisciplinary”	“Statistics”	...
“Data science”	5	10	
...			

Unstructured data

Structured data

- We will mostly assume the use of structured data.

THINGS TO WATCH OUT

- Quantitative vs Categorical Data

	“interdisciplinary”	“Statistics”	Document Category
“Data science”	5	10	1
...			

Document category:
1- Science, 2- Arts, 3- Politics, ...

Cannot treat these values as numbers, for anything other than simple identity testing

- Big data vs little data: Challenges,

The analysis cycle time slows as data size grows

Large data sets are complex to visualize

Simple models do not require massive data to fit or evaluate

Priority: Look for the right data rather than arbitrary big data



Max de Marzi

- Classification vs Regression

SOME TECHNOLOGIES/LIBRARIES WE WILL USE



MATH PRELIMINARIES: PROBABILITY

- Readings for the next few lectures:
 - Ch 5, 6 (Watkins)
 - Ch 1 (Wasserman, for more formal treatment)
 - Ch 1, 2.1 (Skiena, for light motivational reading)

MATH PRELIMINARIES: PROBABILITY

- The *experiment* of tossing two dice: one red, one blue.

Any process in which
possible outcomes can be
identified ahead of time



- **Sample space Ω** : set of all possible outcomes of an experiment
In our experiment, each outcome in the sample:
(Value of red die, value of blue die)
- **Our example**: What is $|\Omega|$?

MATH PRELIMINARIES: PROBABILITY

- Our sample space Ω :

$\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),$
 $(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),$
 $(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}.$



- **Event A :** specified subset of the outcomes of an experiment
- **Ex:** Event A (Sum of dice = 7 or Sum of dice = 11)
 $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), (5, 6), (6, 5)\}$

AXIOMS OF PROBABILITY

- For any event $A \subseteq \Omega$:

$P(A)$: the probability that the event A occurs.

- Axiom-1: For every event A , $P(A) \geq 0$
- Axiom-2: $P(\Omega) = 1$
- Axiom-3: For any sequence of **disjoint** events $A_1, A_2, \dots$

$$P\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} P(A_i)$$

PROBABILITY AND SET THEORY

- Formal mathematical model for events: Set theory

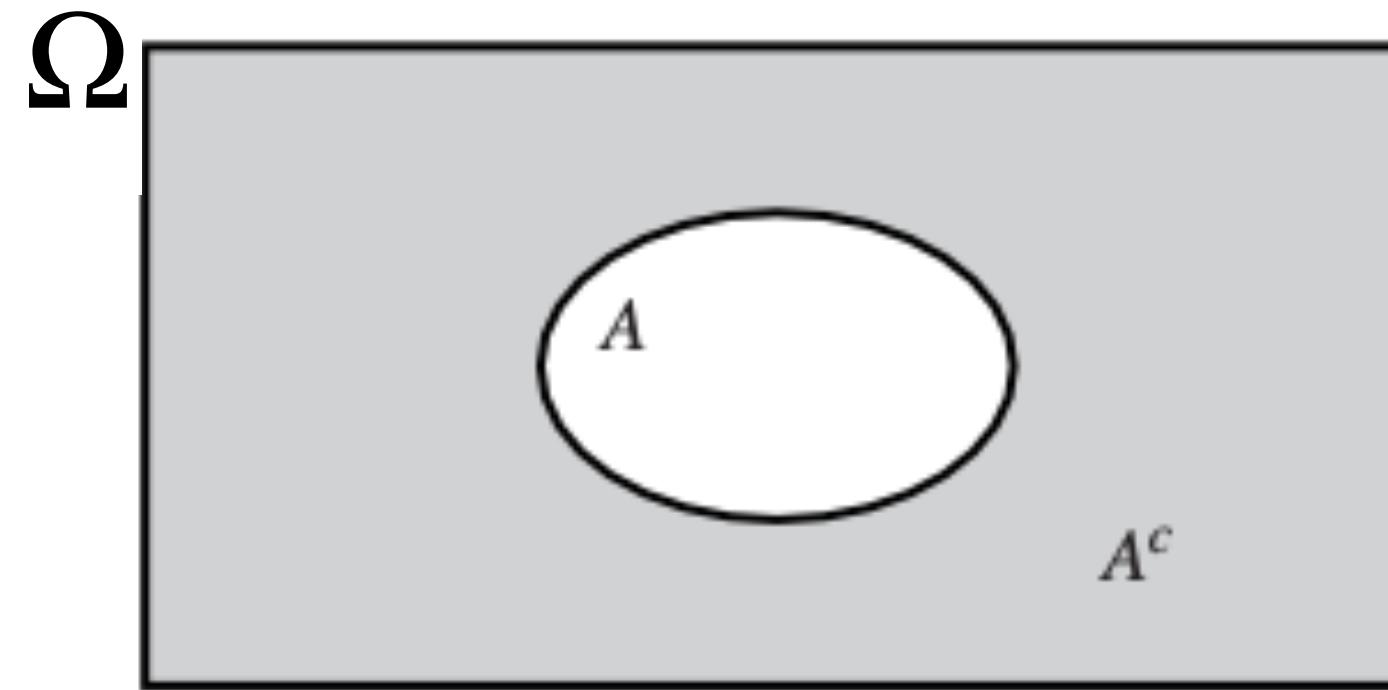
Sample space Ω : a set S

Outcome: an element of S

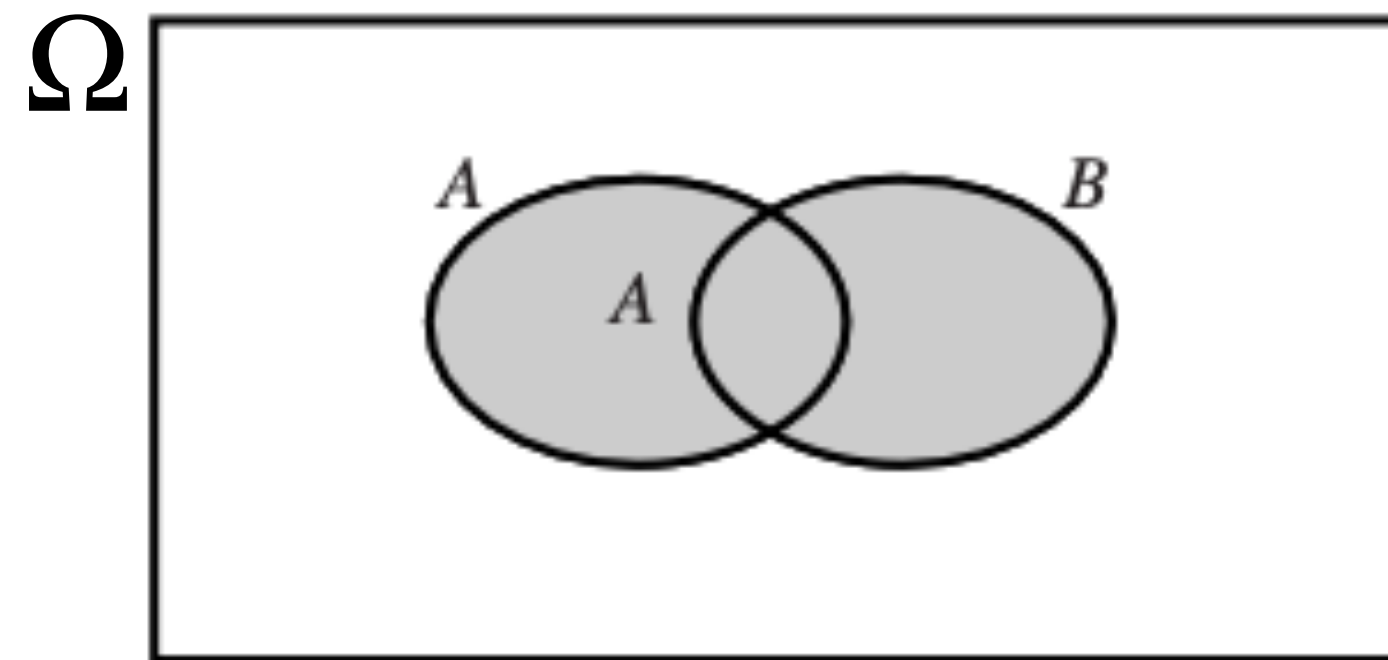
Event: a subset of S

- Further properties of probabilities:
deduce from axioms and set theory

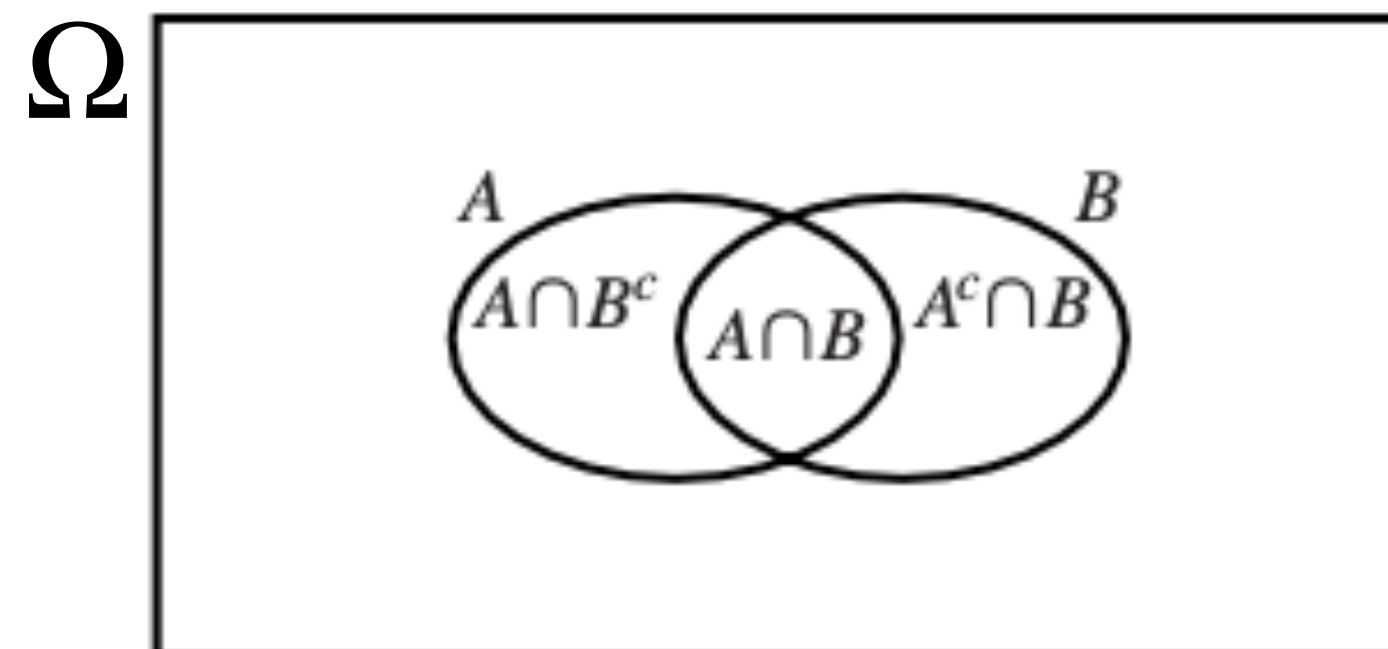
SET THEORY: VERY BRIEF REVIEW



The event A^c



The event $A \cup B$



The partition of $A \cup B$

FURTHER PROPERTIES OF PROBABILITY

- $P(\emptyset) = 0$ ($\emptyset$ refers to an impossible event, e.g. 0 on a die)
- For an event A , $P(A) = 1 - P(A^c)$ (complement)
- $A \subset B \Rightarrow P(A) \leq P(B)$ (monotonicity)
- For any events A, B :
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (inclusion-exclusion)
- ...

SIMPLE SAMPLE SPACE

- **Finite** # of outcomes: $s_1, s_2, \dots, s_n$ and each **equally likely**:

$$\text{Event } A \text{ has } m \text{ outcomes} \Rightarrow P(A) = \frac{m}{n}$$

Ex: Event A : (Sum of the dice = 7 or Sum of the dice = 11)

$$A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1), (5,6), (6,5)\}$$

$$P(A) = \frac{8}{36}$$

FURTHER EXAMPLES

Ex: Disease diagnosis

Patient's symptoms: sore throat, fever

Doctor's diagnosis: viral or bacterial or both

Probability of viral = 0.8, probability of bacterial=0.4



Probability that the patient has both?

$$P(B \cup V) = P(B) + P(V) - P(B \cap V)$$

$$1 = 0.4 + 0.8 - P(B \cap V) \Rightarrow P(B \cap V) = 0.2$$

FURTHER EXAMPLES

Ex: Sampling without replacement

- A president (P) and a treasurer (T) chosen from 20 people including Alice, Bob.
- Probability that Alice is president and Bob is treasurer?

$$\# \text{ of ways to select (P, T): } 20 \times 19 = 380 \quad \Rightarrow \quad \frac{1}{380}$$